

- RAG Learning

2.1 Algebraic Notation & Manipulation

AQA GCSE Mathematics 8300 — Algebra — spec refs A1–A4 — Papers 1 (non-calculator), 2 and 3 (calculator)

2.1.1 Algebraic Notation (A1)

Algebra uses letters to stand for numbers. AQA expects you to read and write algebra using the standard conventions below, and to give answers in their **simplest form** even when the question does not say so.

Instead of...	Write...	Notes
$a \times b$	ab	Leave out the $\times$ sign; letters usually in alphabetical order
$y + y + y$ or $3 \times y$	$3y$	Number first, then the letter
$a \times a$	a^2	" a squared"
$a \times a \times a$	a^3	" a cubed"
$a \times a \times b$	a^2b	Only the a is squared
$a \div b$	$\frac{a}{b}$	Division is written as a fraction
$0.5x$	$\frac{1}{2}x$ or $\frac{x}{2}$	Coefficients written as fractions, not decimals
$3 \times (x + 2)$	$3(x + 2)$	Brackets group terms that are multiplied together

Exam Tip: $2a^2$ and $(2a)^2$ are different: $2a^2 = 2 \times a \times a$ but $(2a)^2 = 2a \times 2a = 4a^2$. The power only applies to what it is directly attached to.

2.1.2 Substitution (A2)

Substituting means replacing each letter with its given value and then working out the answer using the correct order of operations (brackets, indices, division/multiplication, addition/subtraction). You may be given unfamiliar formulae, including scientific ones; all the information you need will be in the question.

Worked Example: Work out the value of $3p^2 - 2q$ when $p = -4$ and $q = 5$.

$$p^2 = (-4)^2 = 16 \text{ (a negative number squared is positive)}$$

$$3 \times 16 - 2 \times 5 = 48 - 10$$

$$= 38$$

Worked Example: The formula $v = u + at$ gives velocity. Find v when $u = 12$, $a = -3$ and $t = 5$.

$$v = 12 + (-3)(5) = 12 - 15$$

$$v = -3$$

Exam Tip: Always put negative values in brackets when you substitute: $x = -3$ in x^2 gives $(-3)^2 = 9$, not -9 . On a calculator type the bracket too.

2.1.3 Vocabulary: Expressions, Equations, Formulae, Identities (A3)

Term: a single number, letter, or product of numbers and letters, e.g. $5x^2$, $-3y$, 7 .

Coefficient: the number in front of a letter, e.g. in $-4x^3$ the coefficient is -4 .

Factor: something that divides exactly into a term or expression, e.g. 3 and x are factors of $3x$; $(x + 2)$ is a factor of $x^2 + 5x + 6$.

Expression: a collection of terms with *no* equals sign, e.g. $4x - 3$.

Equation: two expressions that are equal for particular values of the letter, e.g. $2x + 1 = 9$ (true only when $x = 4$).

Formula: a rule linking two or more variables, e.g. $A = lw$.

Inequality: a statement using $<$, $>$, $\leq$, $\geq$, e.g. $x + 4 < 9$.

Identity: a statement that is true for every value of the letter, written with $\equiv$, e.g. $5(x + 1) \equiv 5x + 5$.

Worked Example: Classify: (A) $4x - 3$ (B) $2x + 1 = 9$ (C) $A = lw$ (D) $5(x + 1) \equiv 5x + 5$.

A has no equals sign $\rightarrow$ **expression**

B is true only when $x = 4 \rightarrow$ **equation**

C links area, length and width $\rightarrow$ **formula**

D is true for all $x \rightarrow$ **identity**

2.1.4 Simplifying and the Laws of Indices (A4)

Collecting like terms

Like terms have exactly the same letters raised to the same powers. Only like terms can be added or subtracted. $3xy$ and $5yx$ are like terms; $4x$ and $4x^2$ are not.

Worked Example: Simplify $7a + 4b - 3a + 2b - b$.

$$a \text{ terms: } 7a - 3a = 4a$$

$$b \text{ terms: } 4b + 2b - b = 5b$$

$$\text{Answer: } 4a + 5b$$

Laws of indices

$$a^m \times a^n = a^{m+n} \quad a^m \div a^n = a^{m-n} \quad (a^m)^n = a^{mn}$$

$$a^0 = 1 \quad a^{-n} = \frac{1}{a^n}$$

To multiply terms, multiply the numbers and add the powers of each letter. To divide, divide the numbers and subtract the powers.

Worked Example: Simplify $(3x^2y^4)^2 \div x^3y$.

$$(3x^2y^4)^2 = 3^2 \times x^4 \times y^8 = 9x^4y^8 \text{ (square the 3 as well!)}$$

$$9x^4y^8 \div x^3y = 9x^{4-3}y^{8-1}$$

$$= 9xy^7$$

Worked Example: Simplify $2x^{-3} \times 5x^5$.

$$\text{Numbers: } 2 \times 5 = 10; \text{ powers: } -3 + 5 = 2$$

$$\text{Answer: } 10x^2$$

2.1.5 Single Brackets and Common Factors (A4)

Expanding a single bracket

Multiply every term inside the bracket by the term outside: $a(b + c) = ab + ac$.

Worked Example: Expand and simplify $4(3x - 2) - 5(x - 3)$.

$$4(3x - 2) = 12x - 8$$

$$-5(x - 3) = -5x + 15 \quad (-5 \times -3 = +15)$$

$$12x - 8 - 5x + 15 = 7x + 7$$

Factorising by taking out common factors

Factorising is the reverse of expanding. Find the **highest common factor (HCF)** of the numbers *and* of the letters, write it outside the bracket, then divide each term by it.

Worked Example: Factorise fully $8x^2y - 12xy^2$.

HCF of 8 and 12 is 4; HCF of x^2y and xy^2 is xy

$$\text{HCF} = 4xy$$

$$8x^2y \div 4xy = 2x \text{ and } 12xy^2 \div 4xy = 3y$$

$$\text{Answer: } 4xy(2x - 3y)$$

Exam Tip: "Factorise fully" means the bracket must have no common factor left. $2xy(4x - 6y)$ is only partly factorised and usually scores 1 mark out of 2.

2.1.6 Expanding Products of Two Binomials (A4)

A **binomial** is an expression with two terms, like $(x + 3)$. To expand $(a + b)(c + d)$, multiply each term in the first bracket by each term in the second, giving four terms, then collect like terms. The grid method keeps this organised.

Grid method: $(2x - 3)(3x + 4)$

	3x	+4	
2x	$6x^2$	$+8x$	$\begin{array}{r} 6x^2 + 8x - 9x - 12 \\ \hline = 6x^2 - x - 12 \end{array}$
-3	$-9x$	-12	

Each cell = row header $\times$ column header**Worked Example:** Expand and simplify $(x + 6)(x - 2)$.

$$x \times x = x^2, \quad x \times (-2) = -2x, \quad 6 \times x = 6x, \quad 6 \times (-2) = -12$$

$$x^2 - 2x + 6x - 12 = x^2 + 4x - 12$$

Special products

$$(a + b)^2 = a^2 + 2ab + b^2 \quad (a - b)^2 = a^2 - 2ab + b^2$$

$$(a + b)(a - b) = a^2 - b^2 \quad (\text{difference of two squares})$$

Worked Example: Kim says $(x + 5)^2 = x^2 + 25$. Show she is wrong.

$$(x + 5)^2 = (x + 5)(x + 5) = x^2 + 5x + 5x + 25 = x^2 + 10x + 25$$

Counter-example: $x = 1$ gives $6^2 = 36$, but $1 + 25 = 26$.**Exam Tip:** Squaring a bracket always means writing it out twice. The most common lost mark at GCSE is forgetting the middle term $2ab$.

2.1.7 Factorising Quadratics $x^2 + bx + c$ (A4)

To factorise $x^2 + bx + c$, find two numbers that **multiply to give** c and **add to give** b . These are the numbers in the brackets.

Worked Example: Factorise $x^2 + 9x + 20$.

Factor pairs of 20: 1×20 , 2×10 , 4×5

$4 + 5 = 9 \checkmark$

Answer: $(x + 4)(x + 5)$

Worked Example: Factorise $x^2 - 2x - 15$.

Need product -15 and sum -2 : -5 and $+3$

Answer: $(x - 5)(x + 3)$ (check: $-5x + 3x = -2x \checkmark$)

Signs in $x^2 + bx + c$	Signs in the brackets	Example
$b > 0, c > 0$	both +	$x^2 + 7x + 12 = (x + 3)(x + 4)$
$b < 0, c > 0$	both -	$x^2 - 7x + 12 = (x - 3)(x - 4)$
$c < 0$	one +, one -	$x^2 - x - 12 = (x - 4)(x + 3)$

Difference of two squares

When there is no x term and both parts are perfect squares separated by a minus sign:

$$a^2 - b^2 = (a + b)(a - b)$$

Worked Example: Factorise $x^2 - 64$.

$$x^2 - 8^2 = (x + 8)(x - 8)$$

Worked Example: Factorise fully $3x^2 - 12x - 36$.

Common factor first: $3(x^2 - 4x - 12)$

Product -12 , sum -4 : -6 and $+2$

Answer: $3(x - 6)(x + 2)$

Exam Tip: Always take out a common factor first. Then check your answer by expanding the brackets — it takes 20 seconds and catches sign errors.

2.1.8 Factorising $ax^2 + bx + c$ and Expanding Three Binomials

Higher

Factorising when $a \neq 1$

1. Work out ac .
2. Find two numbers that multiply to ac and add to b .
3. Split the middle term using these numbers.
4. Factorise in pairs (grouping) and take out the common bracket.

Worked Example: Factorise $6x^2 - 7x - 3$.

$ac = 6 \times (-3) = -18$; numbers with product -18 and sum -7 : -9 and $+2$

$6x^2 - 9x + 2x - 3$

$3x(2x - 3) + 1(2x - 3)$

Answer: $(2x - 3)(3x + 1)$

The difference of two squares also works with coefficients: $9x^2 - 16 = (3x + 4)(3x - 4)$ and $4x^2 - 1 = (2x + 1)(2x - 1)$.

Expanding products of three binomials

Expand any two brackets first, simplify, then multiply the result by the third bracket. You should get up to six terms before collecting.

Worked Example: Expand and simplify $(x + 4)(x - 1)(x + 2)$.

$$(x + 4)(x - 1) = x^2 + 3x - 4$$

$$(x^2 + 3x - 4)(x + 2) = x^3 + 2x^2 + 3x^2 + 6x - 4x - 8$$

$$= x^3 + 5x^2 + 2x - 8$$

Exam Tip: For three brackets, write the six terms in a line before collecting. Check by substituting $x = 1$ into both forms: $(5)(0)(3) = 0$ and $1 + 5 + 2 - 8 = 0$ ✓.

2.1.9 Expressions Involving Surds Higher

Brackets containing surds are expanded in exactly the same way as algebraic brackets. Use $\sqrt{a} \times \sqrt{a} = a$ and $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$, then collect like surds.

Worked Example: Expand and simplify $(\sqrt{5} + 2)(\sqrt{5} - 3)$.

$$\sqrt{5}\sqrt{5} - 3\sqrt{5} + 2\sqrt{5} - 6$$

$$= 5 - \sqrt{5} - 6 = -1 - \sqrt{5}$$

Worked Example: Expand $(2 + \sqrt{5})^2$.

$$(2 + \sqrt{5})(2 + \sqrt{5}) = 4 + 2\sqrt{5} + 2\sqrt{5} + 5$$

$$= 9 + 4\sqrt{5}$$

Note that $(\sqrt{3} + 1)(\sqrt{3} - 1) = 3 - 1 = 2$: the difference of two squares removes the surd entirely.

2.1.10 Algebraic Fractions Higher

Simplifying

Factorise the numerator and denominator *completely*, then cancel any common **bracket** (never cancel individual terms).

Worked Example: Simplify fully $\frac{x^2 - 16}{2x^2 + 9x + 4}$.

Numerator: $(x + 4)(x - 4)$

Denominator: $ac = 8$, numbers 8 and 1: $2x^2 + 8x + x + 4 = 2x(x + 4) + 1(x + 4) = (2x + 1)(x + 4)$

Cancel $(x + 4)$: $\frac{x - 4}{2x + 1}$

Adding and subtracting

Use a common denominator (usually the product of the denominators), adjust each numerator, then combine.

Worked Example: Write $\frac{2}{x + 3} - \frac{1}{x - 2}$ as a single fraction.

Common denominator $(x + 3)(x - 2)$

$$\frac{2(x - 2) - (x + 3)}{(x + 3)(x - 2)}$$

Numerator: $2x - 4 - x - 3 = x - 7$

Answer: $\frac{x - 7}{(x + 3)(x - 2)}$

Multiplying and dividing

Factorise everything, flip the second fraction if dividing, then cancel.

Worked Example: Simplify $\frac{x^2 + 2x - 15}{x^2 - 9} \div \frac{x + 5}{2x + 6}$.

$$\frac{(x + 5)(x - 3)}{(x + 3)(x - 3)} \times \frac{2(x + 3)}{x + 5}$$

Cancel $(x + 5)$, $(x - 3)$ and $(x + 3)$

Answer: 2

Exam Tip: When subtracting, put the second numerator in a bracket: $-(x + 3) = -x - 3$. Missing this sign change is the most common error in algebraic fraction questions.

Quick Reference Summary

Skill	Key rule
Notation	ab , $3y$, a^2b , $\frac{a}{b}$; simplest form always
Substitution	Brackets round negatives; follow order of operations
Vocabulary	Expression, equation, formula, inequality, identity ($\equiv$), term, factor
Indices	Add when multiplying, subtract when dividing, multiply for a power of a power
Single bracket / HCF	$a(b + c) = ab + ac$; take out the HCF of numbers and letters

Two binomials	Four products, then collect; $(a \pm b)^2 = a^2 \pm 2ab + b^2$
$x^2 + bx + c$	Two numbers: multiply to c , add to b
Difference of two squares	$a^2 - b^2 = (a + b)(a - b)$
$ax^2 + bx + c$ Higher	Split middle term using ac , then group
Three binomials Higher	Expand two, then multiply by the third
Algebraic fractions Higher	Factorise fully, cancel brackets; common denominator to add/subtract